Critical literature review

Industrial multimodal monitoring, fusion and reliability

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Prepared for Parth Ravaliya's master's thesis. The original exposé is the controlling statement of the research questions. This review compares it with the proposed research charter and reported evidence; it does not presume a positive adaptive-fusion result.

Principal conclusion

The scientific case for comparing multimodal architectures is sound. The case for declaring the proposed adaptive method novel, superior, or already validated is not. Existing work establishes feature gating, prediction stacking, uncertainty-conditioned fusion and missing-modality training as substantial prior art. Direct welding work now includes dynamic audiovisual fusion. The defensible contribution is presently a bounded comparative and methodological assessment, with an unresolved adaptive-fusion hypothesis.

What the project evidence currently supports

The supplied evidence map reports a modest clean-data advantage for a logistic prediction stacker over the strongest sensor baseline on REASSEMBLE. Its augmentation-matched adaptive pilot reports a small average-precision difference with an interval spanning benefit and harm. Neither result establishes general industrial robustness, calibrated uncertainty, or a novel fusion mechanism. These are document-reported results, not computations reproduced in this review.

How to read this report

Sections 1-3 recover the original questions and audit their alignment. Sections 4-8 synthesize the literature, distinguish architectures, develop conditional hypotheses and assess the contribution. Sections 9-10 document source limitations and reference priorities. The evidence catalogue supplies paper-level problem, dataset, architecture, protocol, result, limitation, DOI/access information and relevance to all four subquestions.

Evidence labels

T = mathematical or methodological theory, with assumptions. P = published empirical finding, limited to its protocol. E = existing project evidence reported in the supplied documents. H = untested project hypothesis or interpretation. A paper's positive result is P, not a theorem; a previous AI review is a project document, not an independent replication.

Search cutoff: 25 October 2026. This is a critical, targeted review with an explicit source-access audit, not an exhaustive systematic review or a novelty certification. No implementation or new experiments are recommended here.

1. The original exposé

Source [P0], pp. 2-3, §3. Line wrapping and PDF hyphenation have been normalized; the wording below is otherwise preserved.

Main research question

> *How can a process-aware multimodal AI architecture be designed and evaluated for robust industrial anomaly detection using robotic welding data as a case study?*

SQ1 - representations

> *Which lightweight or pretrained Transformer-based representation components are suitable for visual, audio, and sensor time-series data in a multimodal industrial anomaly detection pipeline?*

SQ2 - comparative performance

> *How do unimodal baselines, simple concatenation fusion, modular late fusion, and process-aware gated cross-modal fusion compare in terms of predictive performance?*

SQ3 - robustness

> *Can a gated cross-modal fusion architecture improve robustness when one or more modalities are missing, degraded, or unreliable?*

SQ4 - evaluation beyond accuracy

> *How can multimodal architecture performance be evaluated beyond accuracy, including robustness, interpretability, modularity, and practical usability?*

What the original actually commits to

The abstract restricts 'physics-informed' to using process signals, metadata and context; it does not promise differential-equation constraints or a physical simulator. Section 5.2 explicitly permits an RT-DETR pretrained visual backbone without object-localization annotations, using frames or clips. A CNN feature tap is therefore not automatically a breach of the exposé, although it is not evidence that temporal Transformer attention was evaluated.

Sections 5.3-5.5 distinguish embedding concatenation, branch-prediction late fusion and a proposed gate on projected modality representations. Section 5.6 includes temporal shifts as well as removal, noise and visual degradation. Section 5.1 permits supervised quality/defect classification. A positive gated-fusion result or a new algorithm is an expectation to investigate, not a condition for answering the questions.

2. Exposé-to-charter alignment

The charter [P1] is explicitly proposed, not adopted. Its central question concerns quality-conditioned late fusion under partial degradation against an augmentation-matched static comparator, with a clean-performance constraint and group-disjoint evaluation. That is a useful narrowing within SQ3 and part of SQ2. It cannot silently stand in for all four original questions.

Question	Literature alignment	Evidence/charter coverage	Defensible status
SQ1	Visual/video/audio and time-series candidates [R05-R12]; different pretraining objectives and temporal structure.	Frozen RT-DETR CNN features; statistics and scratch-trained PatchTST. Audio mapping unresolved; charter focuses fusion.	Partly addressed for two branches. Not a completed comparison of pretrained three-modality representations.
SQ2	Stacking, feature gates, attention and decision fusion are distinct [R13-R18,R22-R23].	Core F1-F6 compare several families. Pilot interaction head changes the original gate and training recipe.	Comparative answer possible within the documented recipes; no universal architecture ranking.
SQ3	Quality-conditioned weighting and missing-modality resilience have prior art [R18-R21]; temporal alignment is separate [R17].	Partial-degradation pilot: one fold; interval includes zero. Full absence uses the same fallback in both arms.	Adaptive benefit unresolved. Complete absence cannot identify adaptation benefit under a shared fallback.
SQ4	Leakage, shift, calibration, explanation and reproducibility [R24-R34].	Group splits and saved-result audits reported. Threshold utility, calibration, explanation fidelity and deployment costs incomplete.	Methodological assessment is substantial; practical reliability is not yet established.

Corrections to the project narrative

The charter and evidence-map framing relied on an exposé proxy because the original was said not to be located. The original is now available and takes precedence. The earlier FOUR_EYES review [P6] also records that the original was inspected and that the researcher reported supervisor agreement to change datasets. This is documentary evidence of a reported agreement; the underlying correspondence was not inspected here.

The switch from welding to contact-rich assembly changes the application claim, label semantics and deployment population, even if the architecture question survives. The contemplated MSM study is an independently acquired milling dataset with dataset-specific fitting, not zero-shot transfer. Its suitability and any external result remain unresolved [P1,P4,R35].

The original three-modality scope, the current two-modality evaluation and the proposed partial-degradation focus should be stated as separate scopes. Absence of an audio result is not evidence that the public dataset contains no audio.

3. Existing evidence: what it establishes

All values on this and the following page are E: transcribed from EXISTING_EVIDENCE_MAP and its cited audit summaries [P2], not recalculated from predictions or raw data. The five research-direction documents identify baseline commit 3c4fb20bee369c77d16cf5d7b0679605fb349cbc. No repository, model checkpoint or split manifest was independently audited for this review.

Evidence	Reported result and protocol	Permitted inference / limit
E1: PHM	20 runs, three experiments; leave-one-experiment-out provisional image-derived damage regression. MAE: constant 0.804156; sensor-submodality fusion 0.823786; PatchTST 0.941455.	No established improvement over the constant under this protocol. The target is not validated physical wear; only three independent experiment groups. This is not visual-plus-sensor fusion.
E2: branches	REASSEMBLE: 4,530 actions, 509 failures, 148 recordings; five outer/four inner recording folds. AP: sensor statistics + logistic regression 0.429392; frozen CNN visual 0.230658; scratch PatchTST 0.241591; action-only 0.180805.	Sensor statistics are strongest among these recipes. This does not disprove pretrained time-series representations or temporal attention. PatchTST's inner selection chose the largest six-epoch candidate in all folds.
E3: fusion	AP: uniform prediction average 0.445742; logistic probability stacker F2 0.447299; embedding concatenation F3 0.412231; action gate F4 0.443725; quality gate F5 0.444549; combined gate F6 0.441356.	F2 exceeds the sensor baseline by 0.017907; the prior audit reports paired interval [0.008603, 0.028284]. Gates do not lead the table. Complementary information is plausible, but improvement is not isolated from the extra fitting/calibration stage.
E4: feature tap	Matched fixed recipe: F2 AP 0.440764 with CNN features versus 0.424979 with frozen spatial hybrid-encoder features; reported difference -0.015785, interval [-0.024804, -0.005800].	A narrower feature-tap result. Feature width changes (896 vs 768) and downstream capacity confound an 'attention caused the difference' claim. Neither branch models temporal order through attention.

The public REASSEMBLE paper reports 4,551 demonstrations and 4,035 successes [R03]. The project's 4,530/509 cohort therefore excludes 21 demonstrations, including seven failures by arithmetic. This discrepancy is not proof of bias or correctness: exclusion reasons and identifiers were not inspected.

The Intel route was reportedly stopped because of 216 session crossings and an exposed holdout. This is a split-integrity issue, not a published or project finding that welding models perform poorly. Separate development history from architecture performance.

3. Existing evidence: unresolved claims

Evidence	Reported result and protocol	Permitted inference / limit
E5: stress pilot	One outer fold: 753 actions, 77 failures, 26 recordings; six corruption families, three severities and three stochastic repeats. F2 AP: clean 0.307929; strong image noise 0.284640; strong sensor noise 0.169409.	Shows sensitivity of the selected static pipeline on this fold. It does not evaluate adaptive superiority. Repeated corruptions of an action are not new independent actions.
E6: matched pilot	3,777 parent out-of-fold rows, 11,331 augmented rows; 0.50 clean / 0.25 image fault / 0.25 sensor fault weights. AP: static augmented A 0.253996; additive-quality B 0.237360; interaction C 0.257009.	C-A = 0.003012326; reported 2,000-draw recording bootstrap interval [-0.026710962, 0.032165450]. Inconclusive for benefit and harm. One fold and model-capacity differences limit attribution.
E6: clean constraint	Clean C-A difference 0.004771136; reported interval [-0.034258, 0.044544]. Eight observable quality variables, train-only scaling; documents report no labels, fault names or pristine reference features in the predictor.	No demonstrated noninferiority or equivalence. The charter's 0.01 margin was illustrative, not agreed. A non-significant difference is not evidence of an acceptably small loss.
E7/E8: visual tuning	E7 compares unequal training budgets. E8 holds two epochs and 32 frames fixed, uses two seeds and five folds; F2 changes -0.000796 and -0.002822, intervals crossing zero.	Fine-tuning gain is unresolved. Frozen visual recall at threshold 0.5 is reported as zero; tuned recall is 4/509 and 3/509. Ranking quality does not establish useful decisions at that threshold.

What the adaptive pilot actually is

The reported model has logit z = b + a^Tp + c^Tq + d_v^T(p_vq) + d_s^T(p_sq), where p contains branch probabilities and q observable quality descriptors. Its quality-dependent coefficients are unconstrained and q has a direct effect. It is a quality-conditioned logistic interaction model. It is neither a convex trust-weight gate nor the original projected-feature gate. A coefficient cannot be interpreted as a percentage of trust.

A bootstrap of saved predictions is conditional on fitted models, selected recipes and the observed cohort. It does not include the entire training and research-selection process. Nested fitting helps within the specified procedure; it does not undo repeated development decisions made after viewing the same cohort. Later G/C calibration recipes differ from the core nested recipe, so absolute AP values across E3-E8 are not a common leaderboard.

FOUR_EYES [P6] reports conditional reconstruction of saved-array metrics, not a raw-data replication, and describes a source-level PHM padding concern without measuring its output effect. Those statements must retain their scope. The current review has not independently verified either the reported metrics or the code concern.

4. Literature synthesis: industrial monitoring

Task definitions precede architecture comparisons

Industrial monitoring covers several different estimands: retrospective action failure, normal-only anomaly scoring, product-defect localization, process-state classification and early fault warning. Their labels, permissible observation windows and metrics differ. An action classifier using an entire action can support post-action assessment without establishing early warning. Product surface inspection is relevant architectural evidence, but it is not direct evidence for temporal process monitoring.

Stemmer et al. [R01] provide direct welding evidence for a simple audiovisual score combination. Their anomaly models use normal training samples, but fusion selection uses defect-containing validation data. DMAW [R02] provides much closer dynamic-fusion prior art. M3DM [R04] concerns RGB/3D product inspection. These papers support the general rationale for complementary sensing while requiring different conclusions about supervision, timing and generalization.

REASSEMBLE [R03] makes assembly monitoring a credible case study, but its published segmentation and condition-monitoring tasks do not supply a directly comparable baseline for the project's pooled action-failure AP. A dataset paper establishes available data and demonstrated tasks; it does not validate every later cohort construction or target transformation.

Pretraining is an intervention, not an architecture label

A model may have a Transformer architecture and still be trained from scratch. A pretrained model may contribute only its convolutional backbone. A self-supervised learning method may be trained separately on each downstream dataset, whereas a foundation-model checkpoint has already seen a broad pretraining corpus. These distinctions determine what SQ1 actually tests.

ViT [R05] and the video-specific designs [R07-R08] support the feasibility of attention-based visual representations. RT-DETR [R06] supports an efficient detector architecture, but its detection benchmark cannot justify the performance of every extracted feature tap on action failure. The original exposé's explicit allowance of RT-DETR is compatible with the project's backbone reuse; it does not make CNN pooling into a temporal Transformer.

AST [R09] establishes a pretrained spectrogram route for audio. PatchTST [R10] establishes patch-based time-series forecasting with an optional self-supervised route. TS2Vec [R11] supplies a convolutional contrastive alternative; MOMENT [R12] supplies broad time-series pretraining. Their objectives and transfer protocols differ. The project's scratch PatchTST result answers a budgeted recipe comparison, not the suitability of pretrained time-series models as a class.

Suitability is multidimensional

For SQ1, suitability concerns the target, observation timing, data volume, pretraining provenance, inference cost and representation loss. Mean pooling over independent frame features is invariant to frame permutation: it can retain appearance and sampled-state distributions, but cannot itself encode their order. More sampled frames can change coverage without adding temporal reasoning. No reviewed representation paper establishes superiority for this particular industrial cohort.

4. Literature synthesis: fusion and reliability

There is no single ladder from simple to advanced fusion

Stacking [R13], mixture-of-experts routing [R14], multiplicative feature gates [R15], bottleneck attention [R16] and cross-modal sequence attention [R17] solve different problems. Some preserve modular decisions; others create interactions that cannot be reconstructed from branch probabilities alone. A more expressive model can represent useful cross-modal relationships and can also overfit, exploit shortcuts or increase computational cost.

Published findings run in both directions. MBT's mid-layer interaction improves over its late-fusion setting [R16], whereas Wang et al. [R22] and MultiBench [R23] show that additional modalities or model complexity need not improve a particular result. MultiBench calls feature concatenation 'late fusion'; this report reserves 'prediction-level late fusion' for combinations of branch outputs. Architecture descriptions, not labels, govern the comparison.

Quality, confidence and reliability are different quantities

Signal quality describes observable measurement properties, such as clipping, blur or missingness. Predictive uncertainty is a model-derived description of what it does not know. Reliability concerns whether predictions or uncertainty estimates behave as intended on a specified population. A clean-looking sensor can be wrong for a new operating regime; a noisy measurement can still carry the decisive fault cue.

QMF [R18] and trusted multi-view classification [R19] are prior art for linking per-sample evidence or uncertainty to fusion. DMAW directly brings dynamic weighting into welding [R02]. Their existence contradicts a broad 'first quality-aware industrial fusion' claim. They do not determine whether the project's particular observable descriptors improve its classifier.

A quality descriptor may also measure the physical event being classified. For example, an abrupt sensor change can be either a measurement defect or a real abnormal process. A direct q term can improve prediction through that content association even if it does not identify which branch is trustworthy. This is a plausible alternative explanation of any future quality-head gain, not an observed diagnosis of the pilot.

Reliability cannot be inferred from a gate plot

Even normalized gates need not be calibrated probabilities that a modality is correct. Feature scaling, correlated inputs and alternative parameterizations can change apparent attribution. The attention-explanation debate [R30-R31] supports assessing a specific explanatory claim rather than automatically accepting or rejecting all attention visualizations. The pilot's unconstrained interaction coefficients have an even weaker claim to a trust interpretation.

The present evidence therefore supports asking when adaptation is useful. It does not support assuming that adaptation is required, that a learned quality proxy is faithful, or that better predictive ranking proves interpretable reliability.

4. Literature synthesis: robustness and evaluation

Robustness has several non-equivalent meanings

Complete modality absence is a known availability problem. Partial corruption leaves a usable but unreliable stream. Temporal misalignment breaks correspondence. Population shift changes the joint data distribution; a new machine or operator can do so without visible noise. These cases can overlap, but success on one does not establish success on the others.

ModDrop [R21] and EmbraceNet [R20] show resilience mechanisms that do not require the project's quality head. MulT [R17] handles differing sequence rates and unaligned language streams; that does not establish tolerance to arbitrary sensor-clock drift. WILDS [R24] shows why natural domain shifts deserve a separate evidential category from synthetic corruptions. The latest mechanical reconstruction preprint [R36] is relevant to this boundary but was only screened at abstract level.

Evaluation validity limits every architectural inference

The statistical unit in an industrial dataset may be a recording, production run, workpiece, machine, operator or site. Group-disjoint recordings support a narrower claim than unseen factories. Overlapping windows, shared parent runs, post-event information and preprocessing fitted across held-out data can invalidate apparent generalization. Varma and Simon [R25] address selection-induced optimism; Kapoor and Narayanan [R26] illustrate how corrected leakage can reverse substantive model conclusions.

For the project, strict inner-held-out branch predictions are relevant to stacker validity. However, every fitted normalization, feature selector, checkpoint choice, calibration function and meta-model belongs to the stated learning procedure. The review cannot confirm the full provenance chain from the evidence map alone. The map's awareness of leakage is positive documentation, not a substitute for independent artifact inspection.

Ranking, probability quality and decisions answer different questions

AP is informative for a rare positive class [R29], but depends on prevalence and does not prescribe a threshold. Accuracy can hide missed failures. Calibration describes probability-frequency agreement [R27,R34]; shift can break it [R28]. An AP gain is therefore not evidence of calibrated uncertainty or a useful operational alarm. Conversely, a poor result at threshold 0.5 does not prove that the ranking contains no information.

Predictive uncertainty and uncertainty about a comparison must also be separated. A per-example entropy score is not a confidence interval for an AP difference. Correlated folds and repeated corruptions are not independent replications [R33]. The small adaptive-pilot interval analysis is conditional on the saved fits and 26 observed recordings.

Modularity, usability and reproducibility

Separate encoders facilitate replacement, but a fitted stacker can depend on their score scales; modularity is not guaranteed plug-and-play transfer. Practical usability depends on acquisition window, look-ahead, latency, memory, failure handling and decision consequences. Reproducibility requires a traceable data-to-result procedure [R32], beyond a seed or a table. Those dimensions belong to SQ4 even when their current evidential status is 'not established'.

5. Comparison of related methods

The rows describe mechanisms, not a performance ranking. Scores from different papers use different tasks, prevalence, data splits and budgets and must not be numerically ordered.

Family / examples	What is combined	Strength / intended role	Main limitation for this thesis
Unimodal branches	One stream's statistics, images or sequence representations.	Necessary reference for incremental information and cost.	A weak branch may reflect recipe or budget, not an uninformative modality.
Feature concatenation; project F3	[h_v; h_s] followed by a classifier.	Retains embedding information and supports interactions in the head.	Requires paired representations; scale, dimension and head capacity matter.
Static score averaging; Stemmer [R01], project F1	Fixed-weight branch scores or probabilities.	Simple, modular combination.	Cannot change weight per sample; probability and logit averaging differ.
Prediction stacking; Wolpert [R13], project F2	A fitted meta-classifier on held-out branch predictions.	Learns useful combinations and score corrections.	Unconstrained coefficients are not convex trust weights; in-sample meta-data leak.
Feature gate; GMU [R15], original proposal	Projected features modulated by learned gates.	Input-dependent representation selection.	Selection may follow content rather than reliability; requires paired training.
Convex logit gates; project F4-F6	A weighted sum of branch logits with normalized weights.	Explicit sample-dependent decision mixture.	Normalization alone does not calibrate trust or establish robustness.
QMF / TMC [R18-R19]	Uncertainty-weighted outputs / evidence distributions.	Reliability-sensitive decision combination.	Depends on uncertainty being informative under the evaluated shifts.
Cross-attention / MBT / DMAW [R16-R17,R02]	Tokens or features exchange information before the decision.	Models cross-modal interactions; MBT restricts interaction bandwidth.	Different hypothesis class, training demand and compute from a probability stacker.
ModDrop / EmbraceNet [R21,R20]	Dropped modalities / stochastic selection of docked features.	Learns useful predictions with incomplete streams.	Availability resilience does not by itself identify subtle reliability changes.
Pilot interaction head [P1-P2]	p, q and p×q terms in logistic regression.	Quality-conditioned changes to probability coefficients.	Direct q effects, extra capacity and unconstrained signs prevent a pure-gating interpretation.

Training-time gradient balancing, as in [R22] and part of DMAW, changes how branches learn. Inference-time gating changes how their outputs contribute for an individual input. These mechanisms should not be conflated. Likewise, identical fallback rules make total-absence performance a property shared by both pilot arms, not a distinguishing adaptive mechanism.

6. Theoretical foundation and its boundaries

T1. More information can help an ideal predictor

For a fixed population and loss, if the multimodal function class contains every unimodal predictor (by ignoring extra inputs), then its minimum population risk cannot be larger. This follows directly from minimizing over a larger set. It says nothing about finite-sample estimation, optimization, regularization, computation or deployment shift. It therefore cannot justify 'more modalities always improve performance' for fitted models.

T2. When a quality signal can matter

Let L_m be modality m's prediction loss and w_m a sample-dependent weight. The identity E[w_mL_m] = E[w_m]E[L_m] + Cov(w_m,L_m) explains the intuition: weights negatively associated with branch loss can reduce the weighted-loss term, holding their means fixed. The learned q must carry relevant information about conditional branch error beyond what the combiner already knows. A quality label or proxy name alone does not establish this.

QMF's theorem compares generalization-risk upper bounds under conditions including matched expected weights and non-positive weight-loss correlation [R18, Theorem 2]. Ordering two upper bounds does not order the actual risks. Its logistic-loss argument is not an AP guarantee and does not directly cover the project's unconstrained logistic interaction head.

T3. A useful descriptor is not necessarily a reliability descriptor

For the pilot logit, the effective coefficient of p_v is a_v + d_v^Tq, while c^Tq changes the prediction independently of branch probabilities. Thus quality can affect predictions by reweighting branch information, by predicting the label directly, or both. Those pathways are algebraically distinct. A positive aggregate result would not identify which explanation is correct.

T4. Metrics and claims must match

A strictly increasing transform of a binary score preserves ranking, and therefore preserves AP absent changes in ties or numerical handling. It can change calibration and threshold decisions. Brier and log scores address probability forecasts through proper scoring rules [R34]; they are not interchangeable with AP. None of these facts gives a distribution-free guarantee for a deployed industrial model.

T5. An estimand requires a target population

A clean score concerns one data distribution; a corruption-average score concerns a chosen mixture of conditions and severities. Its weights define what is being averaged. An average can improve while an important condition worsens. A clean noninferiority claim additionally requires a justified margin and an interval compatible with that claim. The reported pilot establishes neither noninferiority nor equivalence.

These foundations justify conditional hypotheses about complementary information and informative quality signals. They do not justify an expected winner, a universal reliability mechanism or a claim of algorithmic novelty. The paper-level theoretical and empirical evidence is kept separate in the catalogue.

7. Hypotheses: support and counterevidence

Hypothesis / claim	Supporting rationale	Contradicting or limiting evidence	Current conclusion
H1: combining streams improves the task.	Complementary observations can improve the ideal predictor; direct industrial examples [R01,R04].	Unequal learning/overfitting [R22]; simple/unimodal winners in [R23]; PHM project comparison lacks a gain.	Plausible, task-specific. E3 supports a modest gain for one stacker recipe.
H2: observable q helps under partial degradation.	Conditional-error information can make adaptation useful; QMF and DMAW are precedents [R18,R02].	q may be a content shortcut; corrupted and real deployment distributions differ; E6 interval spans harm and benefit.	Scientifically testable and unconfirmed for the proposed head.
H3: adaptive fusion improves complete absence.	Availability-aware mechanisms can preserve performance [R20-R21].	Pilot arms invoke the same fallback. There is no adaptive difference in this state by construction.	Not an identifiable advantage of this pilot; cannot be supported by shared fallback performance.
H4: Transformer/pretrained representations are superior.	Transferable representations and temporal modeling have published successes [R05-R12].	Architecture is not pretraining; task and budgets differ. E4/E8 do not show a robust advantage.	No general superiority claim. SQ1 is a suitability assessment.
H5: gate weights explain reliability.	Weights expose part of the computational decision.	Attention is not automatically faithful explanation [R30-R31]; direct q and interactions further complicate interpretation.	Descriptive visualization is possible; calibrated or causal trust remains unestablished.
H6: clean performance is preserved.	A model can in principle learn to ignore unhelpful quality variables.	Finite-sample fitting may not do so. Clean pilot interval includes meaningful loss; no adopted margin.	No current noninferiority/equivalence conclusion.
H7: the finding generalizes across industrial processes.	Some sensing and fusion principles are reusable.	Assembly, welding and milling differ in labels, dynamics and acquisition; no MSM result is reported.	External generalization is untested. Dataset-specific refitting would be a separate claim.
H8: the fusion mechanism is novel.	The exact constrained application may be under-characterized.	Stacking, gates, evidential fusion and direct dynamic welding fusion already exist [R02,R13-R21].	Broad novelty is contradicted by prior art; narrower novelty is not established.

'Contradicting' here includes counterexamples to a universal assertion and evidence against the asserted mechanism. It does not mean that a different-domain paper statistically refutes this particular project. Equally, a positive result in another domain does not validate the pilot.

8. Assessment of the proposed contribution

A defensible contribution at present

The current material can support a critical comparative study of selected representation and fusion recipes for industrial action-outcome monitoring, together with an audit of robustness claims and evaluation validity. A negative or inconclusive gated-fusion result is a legitimate answer to SQ2-SQ3. The contribution does not need to be redescribed as a successful novel architecture to be academically meaningful.

The clean REASSEMBLE comparison is the strongest positive project result reported. The adaptive pilot is a narrower, exploratory result. The original exposé is broader than either: its audio, temporal-shift and evaluation-beyond-accuracy components remain only partly covered. A thesis account should preserve these distinctions rather than moving the goalposts after seeing the results.

Defensible gaps, expressed as uncertainties

Gap A - local evidence gap: the supplied work does not establish when observable quality descriptors improve a matched static combiner on group-disjoint industrial action data. This is a project-specific unanswered question; the broader mechanism already has prior art.

Gap B - attribution gap: a gain from a quality-interaction head would not automatically distinguish reliability-sensitive weighting from label-correlated content or extra model capacity. The proposed interpretation is stronger than the presently identified effect.

Gap C - evaluation gap: published robustness results use different corruption, missingness and domain-shift protocols. There is no directly comparable result in the reviewed set that resolves all of the original exposé's modalities, temporal shifts, calibration and practical-utility requirements under the project's exact conditions. This is a gap in the reviewed evidence, not proof that no such study exists.

Gap D - scope gap: assembly failure, welding defect detection and milling anomalies are not interchangeable outcomes. The extent to which any fusion conclusion travels across them remains unknown. The charter's external-data proposal does not itself supply evidence.

Claims that are not currently warranted

The review does not support claims of the first quality-aware industrial fusion method, adaptive superiority, a calibrated trust mechanism, physics-constrained learning, early fault detection, cross-process generalization, or complete fulfillment of the original three-modality programme. It also does not support dismissing Transformers, audio or adaptation as generally ineffective.

Research-question alignment verdict

Retain the original four questions as the organizing framework. Describe the charter as a proposed, narrower question within that framework and the current experiments as partial evidence with specified scope. This is a conclusion about the thesis argument, not an instruction to implement a new method or run another experiment.

9. Missing sources and evidence-quality limits

Project evidence still unavailable to this review

The underlying split manifests, raw cohort construction, prediction arrays, checkpoint provenance, complete training logs and supervisor correspondence were not independently inspected. The evidence map and reviews can report their existence and findings but cannot serve as independent confirmation of themselves. The PHM image proxy's physical validity and the effect of its reported padding issue remain unresolved.

The audio channel-to-recording map, permissible information at prediction time, temporal alignment and outcome-label timing are not established here. REASSEMBLE describes spoken task narration in collection [R03]; using such audio could introduce annotation or outcome cues. This is a plausible leakage pathway requiring source clarification, not a finding that the project's unimplemented audio branch leaked.

Academic access limits

Some sources were inspected through full-text methods/results passages; others only through their publisher or author abstracts. Each catalogue entry states its depth. In particular, TMC's original full methods were not reliably accessible; Wolpert's full experimental details were unavailable; the MSM publisher full text was blocked; recent 2026 reconstruction work was abstract-screened. Those sources are not used for unverified numerical comparisons.

For proceedings without a DOI in the inspected record, a verified arXiv DOI is labelled as a preprint identifier where available. 'No DOI listed/verified' is not a claim that no identifier exists anywhere. Publisher DOIs for several CVPR papers were not independently resolved; the accessible primary proceedings records remain usable citations.

Limits of the literature itself

High scores do not repair unclear independence of train/test samples. For DMAW, the inspected text did not establish run/session-disjoint splitting or a sufficiently explicit held-out corruption protocol. Reported terminology also alternates between laser welding and a pulsed-MIG setup. Its result is important prior art, but its robustness claim is not equivalent to independently validated industrial generalization [R02].

Benchmark samples, metrics and supervision differ. M3DM's product localization, PatchTST's forecast errors and REASSEMBLE's segmentation metrics are not substitutes for action-level failure AP. Positive published results may reflect selection and publication bias. Lack of a reported result or an unlocated paper is not evidence of no effect or of novelty.

Sources identified but not fully assessed

McKinney et al., 'Unsupervised multimodal fusion of in-process sensor data for advanced manufacturing process monitoring' (2025), DOI 10.1016/j.jmsy.2024.12.003; and 'Lazy Fusion of Multimodal Sensors for Cost-Effective Process Monitoring' (2025), DOI 10.1021/acsengineeringau.5c00009, were identified at record/abstract level. They expand the relevant industrial neighbourhood, but dataset, architecture and independence details remain unextracted. Their unreviewed status further limits any novelty conclusion.

The original exposé's broad background references (e.g. the multimodal survey, Hidden Technical Debt and Data Validation for Machine Learning) remain useful context, but they are not replacements for primary mechanism and evaluation evidence. This review does not claim to exhaust all manufacturing domains, uncertainty methods or 2025-2026 foundation-model literature.

10. Search method, provenance and reference priorities

Search and selection

Searches combined the original model/paper names with industrial process monitoring, welding, assembly, milling, multimodal fusion, quality/reliability, missing modality, noise, domain shift, temporal alignment, leakage, calibration, interpretability and reproducibility. Foundational works and recent 2025-2026 records were both sought. Primary publisher/proceedings records, author-hosted papers and arXiv originals were preferred; secondary indexes were used for discovery, not substantive technical conclusions.

Selection was purposive: direct industrial relevance, closeness to the claimed mechanism, or methodological importance to one of the four questions. Reference-following and title/DOI checks were selective. There was no complete Scopus/Web of Science/IEEE corpus export, preregistered screening flow, double screening or exhaustive forward-citation search. No PRISMA counts are claimed. All external searches and access checks were performed on 25 October 2026.

Project-source provenance

The uploaded items were located by metadata, but their text-reading endpoint did not expose readable contents. Matching-name and matching-size local copies in Downloads were read instead. Byte identity with the uploads was not independently established. [P0] is the nine-page MA_Exposé_Ravaliya_Parth.pdf. [P1]-[P5] are RESEARCH_CHARTER, EXISTING_EVIDENCE_MAP, LITERATURE_POSITIONING, OPEN_RESEARCH_DECISIONS and EXECUTIVE_REVIEW in thesis_research_direction, dated 25 October 2026. [P6] is FOUR_EYES_REVIEW_CURRENT; relevant opening findings were inspected, not its entire 125 KB text.

The five research-direction documents were read as project assertions. The original exposé was read in full. No project file was modified. This provenance caveat matters because several project documents disagree about whether the original exposé was available.

References to prioritize in the thesis

For the application and data argument: [R01-R04] and [R35], with MSM access limitations retained. For the actual representation choices: [R05-R12], citing the pretraining and temporal mechanism actually used. For fusion positioning: [R13-R23], especially GMU, MBT, QMF, EmbraceNet and the direct welding precedent DMAW. For defensible evaluation: [R24-R34], especially nested selection, leakage, calibration under shift and the paired explanation debate.

Use the evidence catalogue as a bibliography with access links and a question-level matrix. Entries marked 'limited' or 'screened' are references for background or follow-up reading, not fully verified support for detailed protocol or performance claims. The recent [R36] is a preprint, not a verified peer-reviewed finding.

Review checkpoint

Checked: original question wording and methodological allowances; alignment with the proposed charter; reported project results; primary-source mechanisms and selected result/protocol passages. Still uncertain: raw-data/result reproducibility, missing source details, audio/timing semantics, external validity and adaptive benefit. Defensible conclusion: the comparative thesis direction is justified; the proposed adaptive advantage and any narrow novelty claim remain unresolved.

11. Original-question literature matrix

D = directly addresses the question's mechanism or evaluation issue; I = indirect conceptual/task relevance; - = no substantive answer. D does not mean direct validation on this thesis's data. Detailed relevance and source depth follow in each paper profile.

Original source	SQ1	SQ2	SQ3	SQ4
[R01] Stemmer: welding	D	D	I	D
[R02] DMAW	D	D	D	D
[R03] REASSEMBLE	D	I	I	D
[R04] M3DM	D	D	I	D
[R05] ViT	D	I	-	I
[R06] RT-DETR	D	I	-	D
[R07] Video Swin	D	I	-	I
[R08] TimeSformer	D	I	-	I
[R09] AST	D	I	-	I
[R10] PatchTST	D	I	I	D
[R11] TS2Vec	D	I	I	D
[R12] MOMENT	D	I	I	D
[R13] Stacked Generalization	-	D	I	D
[R14] Mixtures of Local Experts	I	D	I	I
[R15] GMU	I	D	I	I
[R16] MBT	D	D	I	D
[R17] MulT	D	D	D	I
[R18] QMF	I	D	D	D

SQ1: representations. SQ2: comparative fusion performance. SQ3: robustness. SQ4: evaluation beyond accuracy. No single paper settles all four in the original industrial setting.

11. Original-question literature matrix (continued)

D = directly addresses the question's mechanism or evaluation issue; I = indirect conceptual/task relevance; - = no substantive answer. D does not mean direct validation on this thesis's data. Detailed relevance and source depth follow in each paper profile.

Original source	SQ1	SQ2	SQ3	SQ4
[R19] TMC	I	D	D	D
[R20] EmbraceNet	I	D	D	D
[R21] ModDrop	I	D	D	I
[R22] Training Multimodal Networks	I	D	I	D
[R23] MultiBench	D	D	D	D
[R24] WILDS	I	I	D	D
[R25] Nested selection (Varma)	I	D	I	D
[R26] Leakage (Kapoor)	I	D	I	D
[R27] Calibration (Guo)	I	I	I	D
[R28] Uncertainty under shift	I	I	D	D
[R29] Precision-recall (Saito)	-	I	I	D
[R30] Attention is not Explanation	-	I	-	D
[R31] Attention is not not Explanation	-	I	-	D
[R32] Reproducibility (Pineau)	I	I	I	D
[R33] CV variance (Bengio)	I	D	I	D
[R34] Proper scoring rules	-	I	I	D
[R35] MSM milling dataset	I	I	I	D
[R36] Cross-modal reconstruction	I	D	D	D

SQ1: representations. SQ2: comparative fusion performance. SQ3: robustness. SQ4: evaluation beyond accuracy. No single paper settles all four in the original industrial setting.

12. Paper evidence catalogue

FT means relevant full-text passages were inspected, not independent reproduction. A means abstract/record level. Limited and screened entries retain unresolved fields. Numerical values belong only to the cited protocol.

[R01] Unsupervised Welding Defect Detection Using Audio and Video

Stemmer, G., et al. (2024). arXiv preprint.

DOI: 10.48550/arXiv.2409.02290 (preprint)

Source: https://arxiv.org/html/2409.02290v1

Access: FT: §§2-4 and result tables

Problem / data: Normal-trained weld-defect scoring. More than 4,000 welding samples; good versus defective welds across 11 defect types (Table 4's split totals 4,040).

Architecture / fusion: Audio spectrogram autoencoder; frozen SlowFast video features plus an autoencoder; standardized anomaly scores combined by a fixed weight.

Training / evaluation: 576 good training samples; validation 122 good/1,610 defective; test 121 good/1,611 defective. Validation selects fusion weights.

Relevant result: Fusion AUROC 0.9178 versus video 0.8977.

Limits / thesis relevance: Defect labels enter model selection. Whole-sample aggregation and centered video windows do not establish early warning; run independence is not established here.

SQ1 D: frozen visual/audio route | SQ2 D: static late fusion | SQ3 I: no missingness proof | SQ4 D: supervision and timing

[R02] DMAW: A Dynamic Multimodal Measurement Fusion Network With Attention for Reliable Welding Process Monitoring Under Harsh Industrial Environments

Fan, J., et al. (2026). IEEE Sensors Journal, 26(2), 2347-2358.

DOI: 10.1109/JSEN.2025.3634585

Source: https://publikationen.bibliothek.kit.edu/1000189165/177522679

Access: FT: author postprint §§II-III

Problem / data: Welding-state classification amid unreliable audiovisual sensing. 13,000 reported synchronized image-audio samples; four penetration/state classes.

Architecture / fusion: Self-distilled ViT-Small features, cross-attention, dynamic prediction weights and training-time gradient balancing.

Training / evaluation: Twenty-frame groups; Adam, learning rate 2e-5, batch 32, cross-entropy; unimodal/fusion comparisons and ablations.

Relevant result: Reported accuracy 96.1%.

Limits / thesis relevance: Run-disjoint split and held-out degradation design not established in inspected text; 26.2 ms model timing excludes acquisition. Important close prior art, not a verified deployment result.

SQ1 D: learned representations | SQ2 D: dynamic hybrid fusion | SQ3 D: claimed noise robustness | SQ4 D: protocol/latency limits

12. Paper evidence catalogue (continued)

[R03] REASSEMBLE: A Multimodal Dataset for Contact-rich Robotic Assembly and Disassembly

Sliwowski, D., et al. (2025). Author paper, arXiv v2.

DOI: 10.48550/arXiv.2502.05086 (paper); 10.48436/0ewrv-8cb44 (dataset)

Source: https://arxiv.org/html/2502.05086v2

Access: FT: collection, benchmarks and limitations

Problem / data: Contact-rich assembly/disassembly learning and monitoring. 4,551 demonstrations, 4,035 successes, four actions, 17 objects; RGB, events, force/torque, microphones and robot state.

Architecture / fusion: Multiple task baselines: DiffAct segmentation, policy learning and ConditionNET monitoring; not one common fusion architecture.

Training / evaluation: Task-specific training/evaluation, including monitoring with ground-truth action boundaries.

Relevant result: DiffAct frame accuracy 61.5%; condition prediction is markedly stronger for effects than preconditions.

Limits / thesis relevance: Published tasks are not the project's action-failure AP. Collection, exclusions and annotation information require separate scrutiny.

SQ1 D: available streams | SQ2 I: task-specific baselines | SQ3 I: no matched corruption test | SQ4 D: labels and timing

[R04] Multimodal Industrial Anomaly Detection via Hybrid Fusion (M3DM)

Wang, Y., Peng, J., Zhang, J., Yi, R., Wang, Y., & Wang, C. (2023). CVPR.

DOI: 10.48550/arXiv.2303.00601 (preprint; publisher DOI not independently resolved)

Source: https://arxiv.org/html/2303.00601v2

Access: FT: method and experimental sections

Problem / data: Normal-only RGB/3D product-defect detection and localization. MVTec-3D AD, ten object categories.

Architecture / fusion: Frozen DINO image and Point-MAE point-cloud encoders; spatial alignment, patch contrastive fusion, three memory banks and one-class decision fusion.

Training / evaluation: Normal training data; benchmark test anomalies; modality/fusion ablations.

Relevant result: Mean image AUROC 0.945 and AUPRO 0.964.

Limits / thesis relevance: Spatial product inspection, not process sequences. Alignment and normal-memory assumptions limit transfer; missing or drifting process sensors are not established by these scores.

SQ1 D: pretrained encoders | SQ2 D: hybrid feature/decision fusion | SQ3 I: anomaly task, not dropout proof | SQ4 D: distinct metrics/task

12. Paper evidence catalogue (continued)

[R05] An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

Dosovitskiy, A., et al. (2021). ICLR.

DOI: 10.48550/arXiv.2010.11929 (preprint)

Source: https://arxiv.org/html/2010.11929v2

Access: FT: architecture, transfer protocol and comparisons

Problem / data: Can a pure Transformer learn competitive image representations? ImageNet, CIFAR and transfer benchmarks including VTAB.

Architecture / fusion: Image patches become tokens in a Transformer encoder; class-token classification, without cross-modal fusion.

Training / evaluation: Large-scale supervised pretraining, including ImageNet-21k/JFT, followed by downstream fine-tuning.

Relevant result: Large-data pretraining enables strong transfer; smaller-data regimes expose weaker inductive bias.

Limits / thesis relevance: Scale and pretraining are part of the result. It supplies no industrial failure, temporal reasoning or modality-reliability guarantee.

SQ1 D: visual representation | SQ2 I: branch only | SQ3 -: no fusion robustness test | SQ4 I: provenance and compute

[R06] DETRs Beat YOLOs on Real-time Object Detection (RT-DETR)

Zhao, Y., et al. (2024). CVPR.

DOI: 10.48550/arXiv.2304.08069 (preprint)

Source: https://openaccess.thecvf.com/content/CVPR2024/html/Zhao_DETRs_Beat_YOLOs_on_Real-time_Object_Detection_CVPR_2024_paper.html

Access: A+: primary record and author architecture description

Problem / data: End-to-end object detection with competitive speed. COCO detection; additional Objects365 pretraining results.

Architecture / fusion: CNN backbone, efficient hybrid spatial encoder and Transformer decoder; uncertainty-minimal query selection.

Training / evaluation: Supervised box-detection training; COCO AP and T4 GPU throughput comparisons.

Relevant result: R50 reports 53.1 detection AP and 108 FPS.

Limits / thesis relevance: These are detector metrics and hardware timings, not action-failure AP or frozen-backbone latency. The isolated CNN tap contains no Transformer attention.

SQ1 D: permitted backbone route | SQ2 I: branch extraction | SQ3 -: no modality fault test | SQ4 D: naming/metric discipline

12. Paper evidence catalogue (continued)

[R07] Video Swin Transformer

Liu, Z., et al. (2022). CVPR.

DOI: 10.48550/arXiv.2106.13230 (preprint)

Source: https://arxiv.org/abs/2106.13230

Access: A+: proceedings and author records; full PDF access blocked

Problem / data: Efficient video recognition with local spatiotemporal attention. Kinetics-400/600 and Something-Something v2 action recognition.

Architecture / fusion: Hierarchical shifted 3D windows; extends image Swin with temporal attention, no multimodal fusion.

Training / evaluation: Image-pretrained initialization and video fine-tuning; benchmark action-classification evaluation.

Relevant result: Published Kinetics-400 top-1 accuracy 84.9; Something-Something v2 69.6.

Limits / thesis relevance: Video benchmark success does not establish suitability for scarce industrial failures; image pretraining, sampling and compute matter.

SQ1 D: genuine video model | SQ2 I: possible branch | SQ3 -: no process-sensor robustness | SQ4 I: feasibility tradeoff

[R08] Is Space-Time Attention All You Need for Video Understanding? (TimeSformer)

Bertasius, G., Wang, H., & Torresani, L. (2021). ICML, PMLR 139, 813-824.

DOI: No DOI listed in inspected PMLR record.

Source: https://proceedings.mlr.press/v139/bertasius21a.html

Access: A: primary proceedings record

Problem / data: Convolution-free video classification. Kinetics-400/600 and action-recognition benchmarks.

Architecture / fusion: Patch tokens with divided temporal and spatial attention within blocks; no cross-modal fusion.

Training / evaluation: Comparison of attention designs and video-recognition evaluation; detailed training settings were not extracted here.

Relevant result: Divided attention is the best tested design in the paper's comparisons; long clips are supported.

Limits / thesis relevance: Record-level review cannot independently audit exact protocols. It is a temporal-model precedent, not evidence that pooled image features or industrial gates are effective.

SQ1 D: temporal visual candidate | SQ2 I: branch only | SQ3 -: no modality quality test | SQ4 I: compute/temporal scope

12. Paper evidence catalogue (continued)

[R09] AST: Audio Spectrogram Transformer

Gong, Y., Chung, Y.-A., & Glass, J. (2021). Interspeech, 571-575.

DOI: 10.21437/Interspeech.2021-698

Source: https://www.isca-archive.org/interspeech_2021/gong21b_interspeech.html

Access: FT-selected: primary paper/record

Problem / data: Audio classification without convolutional feature extraction. AudioSet tagging, ESC-50 environmental sounds and Speech Commands V2.

Architecture / fusion: Log-mel spectrogram patches, Transformer encoder and classification head; image-pretrained initialization.

Training / evaluation: Audio training/fine-tuning and dataset-specific benchmark evaluation.

Relevant result: AudioSet mAP 0.485; ESC-50 accuracy 95.6%; Speech Commands V2 98.1%.

Limits / thesis relevance: Natural audio transfer does not validate industrial narration-free signals, alignment, early detection or missing-modal robustness.

SQ1 D: audio candidate | SQ2 I: audio branch | SQ3 -: no fusion fault evidence | SQ4 I: checkpoint/task provenance

[R10] A Time Series is Worth 64 Words: Long-term Forecasting with Transformers (PatchTST)

Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). ICLR.

DOI: 10.48550/arXiv.2211.14730 (preprint)

Source: https://arxiv.org/html/2211.14730v2

Access: FT: architecture and evaluation

Problem / data: Long-horizon time-series forecasting. Weather, traffic, electricity, ILI and four ETT datasets.

Architecture / fusion: Temporal patches with channel-independent Transformer processing and shared weights; no cross-modal fusion.

Training / evaluation: Chronological forecasting benchmarks with MSE/MAE; supervised and masked self-supervised variants.

Relevant result: Competitive/improved forecasting across the studied horizons; patching reduces token cost.

Limits / thesis relevance: Forecasting is not failure classification. Channel independence does not directly learn cross-channel attention. Using the architecture from scratch is not evidence of pretrained transfer.

SQ1 D: sensor candidate | SQ2 I: branch design | SQ3 I: representation, not fusion proof | SQ4 D: objective/protocol mismatch

12. Paper evidence catalogue (continued)

[R11] TS2Vec: Towards Universal Representation of Time Series

Yue, Z., Wang, Y., Duan, J., Yang, T., Huang, C., Tong, Y., & Xu, B. (2022). AAAI.

DOI: 10.1609/aaai.v36i8.20881

Source: https://arxiv.org/html/2106.10466v4

Access: FT: method, classification and appendix protocol

Problem / data: Learn representations usable across temporal tasks. Main classification comparison: 125 UCR and 29 UEA datasets; also forecasting/anomaly benchmarks.

Architecture / fusion: Dilated convolutional encoder with hierarchical temporal/instance contrastive learning; no Transformer or fusion.

Training / evaluation: Unlabelled representation learning, max-pooled features and RBF-SVM classification; dataset-specific benchmarks.

Relevant result: Reported average classification improvements 2.4 and 3.0 percentage points over the compared unsupervised methods.

Limits / thesis relevance: Mostly per-dataset learning, not automatically a universal pretrained checkpoint. Its anomaly evaluation includes segment adjustment, so scores are not directly transferable.

SQ1 D: non-Transformer alternative | SQ2 I: sensor branch | SQ3 I: missing-value analysis | SQ4 D: evaluation semantics

[R12] MOMENT: A Family of Open Time-series Foundation Models

Goswami, M., et al. (2024). ICML, PMLR 235.

DOI: 10.48550/arXiv.2402.03885 (preprint)

Source: https://arxiv.org/html/2402.03885v2

Access: FT: pretraining and classification/anomaly appendices

Problem / data: Reusable time-series representations across tasks. Time-series pretraining collection; classification on 95 selected UCR datasets, with further UEA results; anomaly/forecast/imputation tasks.

Architecture / fusion: Patch-based T5-style Transformer encoder and masked reconstruction; task heads or downstream classifiers.

Training / evaluation: Broad pretraining; frozen representations with RBF-SVM classification and other task-specific evaluations.

Relevant result: Shows useful transfer without fitting the encoder separately to each classification dataset.

Limits / thesis relevance: Restricted dataset subsets and partly literature-sourced baselines limit equivalence of comparisons; anomaly threshold selection is outside its stated scope.

SQ1 D: genuinely pretrained sensor route | SQ2 I: candidate branch | SQ3 I: reconstruction, not fusion guarantee | SQ4 D: corpus/threshold limits

12. Paper evidence catalogue (continued)

[R13] Stacked Generalization

Wolpert, D. H. (1992). Neural Networks, 5(2), 241-259.

DOI: 10.1016/S0893-6080(05)80023-1

Source: https://www.sciencedirect.com/science/article/pii/S0893608005800231

Access: A: publisher abstract

Problem / data: Reduce generalization error by learning how base predictors behave. General supervised prediction; detailed original empirical datasets were not verified here.

Architecture / fusion: A second-level learner uses base-model predictions rather than the original feature space.

Training / evaluation: Base predictions for the second level are generated on cases excluded from the corresponding base fit.

Relevant result: Establishes the stacking construction; no unverified numerical gain is quoted.

Limits / thesis relevance: Full experimental protocol unavailable in this review. Stacking is established prior art, but this source alone does not prove that the project's implementation is free of leakage.

SQ1 -: not encoder selection | SQ2 D: prediction-level fusion | SQ3 I: no automatic robustness | SQ4 D: meta-data provenance

[R14] Adaptive Mixtures of Local Experts

Jacobs, R. A., Jordan, M. I., Nowlan, S. J., & Hinton, G. E. (1991). Neural Computation, 3(1), 79-87.

DOI: 10.1162/neco.1991.3.1.79

Source: https://www.cs.toronto.edu/~hinton/absps/jjnh91.pdf

Access: FT-limited: scanned author paper and indexed passages

Problem / data: Reduce interference by assigning cases to specialized experts. Vowel discrimination from acoustic features.

Architecture / fusion: Simple expert networks and an input-dependent gating network; competitive specialization.

Training / evaluation: Speaker-separated training/test; repeated simulations compare expert mixtures and backpropagation networks.

Relevant result: Reported test correctness is 90% for compared models; mixtures reach the training criterion faster (Table 1).

Limits / thesis relevance: Not multimodal sensor-quality fusion; no missingness guarantee. Exact hyperparameter extraction from the scanned paper was incomplete.

SQ1 I: modular branches | SQ2 D: gating foundation | SQ3 I: conditional routing rationale | SQ4 I: efficiency interpretation

12. Paper evidence catalogue (continued)

[R15] Gated Multimodal Units for Information Fusion

Arevalo, J., Solorio, T., Montes-y-Gómez, M., & González, F. A. (2017). Author/workshop paper.

DOI: 10.48550/arXiv.1702.01992 (preprint)

Source: https://arxiv.org/html/1702.01992v1

Access: FT: equations, split and Table 2

Problem / data: Learn how visual and textual representations contribute to classification. MM-IMDb posters and plots; multilabel movie genres.

Architecture / fusion: Learned sigmoid multiplicative gate combines transformed modality embeddings.

Training / evaluation: Stratified train/development/test: 15,552/2,608/7,799; supervised optimization and multiple F1 averages.

Relevant result: Macro-F1 0.541 versus concatenation 0.521; a synthetic noisy-input example also illustrates gate selection.

Limits / thesis relevance: Genre classification is not industrial reliability. A feature gate may select semantic content; clean performance does not prove robustness to sensor faults.

SQ1 I: representation interface | SQ2 D: feature-level gate | SQ3 I: toy noise, not industrial faults | SQ4 I: gate meaning limited

[R16] Attention Bottlenecks for Multimodal Fusion

Nagrani, A., Yang, S., Arnab, A., Jansen, A., Schmid, C., & Sun, C. (2021). NeurIPS.

DOI: 10.48550/arXiv.2107.00135 (preprint)

Source: https://proceedings.neurips.cc/paper/2021/file/76ba9f564ebbc35b1014ac498fafadd0-Paper.pdf

Access: FT: architecture, §4 and ablations

Problem / data: Efficient interaction of audiovisual tokens. AudioSet, VGGSound and EPIC-Kitchens recognition.

Architecture / fusion: ImageNet-21k-initialized ViT branches exchange information through four bottleneck tokens.

Training / evaluation: Supervised training; fusion-layer ablations on mini-AudioSet with three runs; multiple test crops.

Relevant result: Mid-layer fusion beats the tested late setting; early bottlenecks improve over full cross-attention by over two mAP points at under half its compute.

Limits / thesis relevance: Not a quality-aware gate. Benchmark ablation/selection on evaluation data is not an untouched industrial validation protocol.

SQ1 D: audiovisual Transformers | SQ2 D: intermediate fusion | SQ3 I: no process-fault guarantee | SQ4 D: compute/selection

12. Paper evidence catalogue (continued)

[R17] Multimodal Transformer for Unaligned Multimodal Language Sequences (MulT)

Tsai, Y.-H. H., et al. (2019). ACL, 6558-6569.

DOI: 10.18653/v1/P19-1656

Source: https://aclanthology.org/P19-1656.pdf

Access: FT: method, experiments and hyperparameters

Problem / data: Fuse sequences with unequal sampling and cross-time dependencies. CMU-MOSI/MOSEI sentiment and IEMOCAP emotion; language, vision and audio.

Architecture / fusion: Directional pairwise cross-attention, temporal projections and subsequent self-attention/classification.

Training / evaluation: Aligned and unaligned benchmarks; validation-based grid choices and learning-rate decay.

Relevant result: Unaligned performance drops less than several tested alternatives; strong task results.

Limits / thesis relevance: Different sampling rates are not arbitrary clock drift. Language-dominant tasks and full-utterance context do not establish causal online industrial monitoring.

SQ1 D: temporal representations | SQ2 D: cross-attention family | SQ3 D: alignment, with caveat | SQ4 I: observation window

[R18] Provable Dynamic Fusion for Low-Quality Multimodal Data (QMF)

Zhang, Q., et al. (2023). ICML, PMLR 202.

DOI: 10.48550/arXiv.2306.02050 (preprint)

Source: https://proceedings.mlr.press/v202/zhang23ar/zhang23ar.pdf

Access: FT: Theorems 1-2, method and tables

Problem / data: Fusion with varying per-sample modality reliability. NYU Depth V2, SUN RGB-D, UPMC Food-101 and MVSA classification.

Architecture / fusion: Energy-based uncertainty-weighted late fusion with correlation-oriented regularization.

Training / evaluation: Pretrained visual/text branches; synthetic corruptions; repeated runs.

Relevant result: NYU high-noise accuracy 55.60 versus late fusion 51.99; clean 70.09 versus TMC 71.06.

Limits / thesis relevance: Conditional risk-bound result, not an AP or universal superiority theorem; not the project's observable-q interaction model.

SQ1 I: pretrained branches | SQ2 D: quality-aware late fusion | SQ3 D: synthetic corruption | SQ4 D: theory scope

12. Paper evidence catalogue (continued)

[R19] Trusted Multi-View Classification (TMC)

Han, Z., Zhang, C., Fu, H., & Zhou, J. T. (2021). ICLR.

DOI: 10.48550/arXiv.2102.02051 (preprint)

Source: https://arxiv.org/abs/2102.02051

Access: A+: author abstract; full original methods access blocked

Problem / data: Combine views while expressing uncertainty about predictions. Multi-view classification benchmarks; original per-dataset splits and numerical tables not independently extracted.

Architecture / fusion: Per-view Dirichlet evidence combined through Dempster-Shafer reasoning.

Training / evaluation: Joint evidential learning; the authors report classification, robustness and out-of-distribution evaluations.

Relevant result: Positive reliability/robustness findings are claimed; no unchecked numerical result is reproduced.

Limits / thesis relevance: An evidential output is not by definition calibrated. The later TPAMI extension has DOI 10.1109/TPAMI.2022.3171983 and must not be conflated with the 2021 protocol.

SQ1 I: per-view encoders | SQ2 D: evidence-level fusion | SQ3 D: uncertainty-conditioned premise | SQ4 D: calibration caveat

[R20] EmbraceNet: A Robust Deep Learning Architecture for Multimodal Classification

Choi, J.-H., & Lee, J.-S. (2019). Information Fusion, 51, 259-270.

DOI: 10.1016/j.inffus.2019.02.010

Source: https://arxiv.org/html/1904.09078v1

Access: FT: mechanism and experimental settings

Problem / data: Classification despite incomplete multimodal inputs. Gas identification and OPPORTUNITY activity recognition.

Architecture / fusion: Docking layers map modalities to a common width; stochastic feature selection and availability-dependent renormalization.

Training / evaluation: Gas training at locations 1/3/5, validation at 2, test at 4; modality-drop and block-loss evaluations.

Relevant result: Reported resilience under missing modalities and partial loss relative to tested fusion alternatives.

Limits / thesis relevance: Stochastic embracement is not an observable-quality gate. Dataset-specific missingness results do not establish new-machine industrial robustness.

SQ1 I: modular representation interfaces | SQ2 D: stochastic feature fusion | SQ3 D: missing/partial signals | SQ4 D: split-specific scope

12. Paper evidence catalogue (continued)

[R21] ModDrop: Adaptive Multi-Modal Gesture Recognition

Neverova, N., Wolf, C., Taylor, G. W., & Nebout, F. (2016; online 2015). IEEE TPAMI, 38(8), 1692-1706.

DOI: 10.1109/TPAMI.2015.2461544

Source: https://ieeexplore.ieee.org/abstract/document/7169562/

Access: A+: publisher method summary; detailed tables not independently extracted

Problem / data: Gesture detection/localization with incomplete channels. ChaLearn 2014 Looking at People; multimodal visual/motion streams and an audio extension.

Architecture / fusion: Multiscale modality branches with gradual fusion and random modality dropping during training.

Training / evaluation: Challenge evaluation and missing/noisy-channel tests reported.

Relevant result: The system placed first of 17 challenge teams; authors report improved missing-channel resilience.

Limits / thesis relevance: Competition rank concerns the whole system, not isolated modality dropout. It supports training-based robustness as prior art, not a universal effect size.

SQ1 I: branch initialization | SQ2 D: gradual fusion | SQ3 D: modality-drop training | SQ4 I: attribution of system gains

[R22] What Makes Training Multi-Modal Classification Networks Hard?

Wang, W., Tran, D., & Feiszli, M. (2020). CVPR.

DOI: 10.48550/arXiv.1905.12681 (preprint)

Source: https://arxiv.org/html/1905.12681v5

Access: FT: training analysis and results

Problem / data: Why adding modalities can worsen learned classification. Kinetics, mini-Sports, mini-AudioSet and EPIC-Kitchens audiovisual tasks.

Architecture / fusion: Video/audio networks with gradient blending: loss weights respond to overfitting/generalization behaviour.

Training / evaluation: Joint-training comparisons and ablations across architectures and datasets.

Relevant result: Naive fusion can underperform a branch; a scratch Kinetics comparison improves from 77.8 to 79.1 top-1 with the proposed training.

Limits / thesis relevance: Training-time weighting is not inference quality gating. These neural optimization findings do not diagnose a frozen-feature logistic pilot.

SQ1 I: branch learning | SQ2 D: multimodal vs unimodal | SQ3 I: optimization, not sensor-fault proof | SQ4 D: matched-comparison logic

12. Paper evidence catalogue (continued)

[R23] MultiBench: Multiscale Benchmarks for Multimodal Representation Learning

Liang, P. P., et al. (2021). NeurIPS Datasets and Benchmarks.

DOI: 10.48550/arXiv.2107.07502 (preprint)

Source: https://arxiv.org/html/2107.07502v2

Access: FT: task protocols, terminology and robustness analyses

Problem / data: Compare multimodal methods across tasks, complexity and imperfections. 15 datasets, ten modalities and 20 tasks across several domains.

Architecture / fusion: Unified implementations of unimodal, concatenation, interaction and Transformer families.

Training / evaluation: Dataset-specific validation and repeated evaluations; corruption curves and complexity measurements.

Relevant result: Simple approaches are competitive; one modality can dominate, and rankings vary by task.

Limits / thesis relevance: Its 'late fusion' includes embedding concatenation. Some robotics results use best validation performance because no unique test split existed. It is not a single cleanly comparable industrial trial.

SQ1 D: heterogeneous encoders | SQ2 D: comparative taxonomy | SQ3 D: imperfections | SQ4 D: benchmarking limits

[R24] WILDS: A Benchmark of in-the-Wild Distribution Shifts

Koh, P. W., et al. (2021). ICML, PMLR 139.

DOI: No DOI listed in inspected proceedings record.

Source: https://proceedings.mlr.press/v139/koh21a/koh21a.pdf

Access: FT-selected: benchmark design and main findings

Problem / data: Evaluate naturally occurring distribution shifts. Ten datasets, including hospital, camera-trap, geographic and temporal shifts.

Architecture / fusion: Dataset-specific predictive architectures; empirical-risk and domain-robust training comparisons, not one fusion method.

Training / evaluation: Explicit in-distribution/out-of-distribution domains and standardized evaluation.

Relevant result: Substantial out-of-distribution performance gaps remain for standard and tested robust methods.

Limits / thesis relevance: Not an industrial multimodal corruption benchmark. Domain-generalization and subpopulation-shift tasks have different targets; synthetic noise cannot substitute for these evidence categories.

SQ1 I: transfer suitability | SQ2 I: comparison discipline | SQ3 D: natural shift | SQ4 D: domain-based protocols

12. Paper evidence catalogue (continued)

[R25] Bias in Error Estimation When Using Cross-validation for Model Selection

Varma, S., & Simon, R. (2006). BMC Bioinformatics, 7, 91.

DOI: 10.1186/1471-2105-7-91

Source: https://link.springer.com/article/10.1186/1471-2105-7-91

Access: FT: simulation protocol and findings

Problem / data: Bias from reusing selection CV as a performance estimate. Simulated null/non-null high-dimensional classification datasets.

Architecture / fusion: Shrunken centroids and SVM; no multimodal architecture.

Training / evaluation: 10-fold/leave-one-out tuning compared with independent tests and nested CV.

Relevant result: Under null data, optimized estimates below 30% error occurred in 18.5% and 38% of simulations, while independent performance was chance.

Limits / thesis relevance: Specific simulations quantify optimism, not the project's error. Nesting addresses the specified algorithm, not unrestricted human reuse of the final evaluation.

SQ1 I: model selection | SQ2 D: fair fusion comparison | SQ3 I: robust-model selection | SQ4 D: leakage/selection

[R26] Leakage and the Reproducibility Crisis in Machine-learning-based Science

Kapoor, S., & Narayanan, A. (2023). Patterns, 4(9), 100804.

DOI: 10.1016/j.patter.2023.100804

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC10499856/

Access: FT: taxonomy and original case-study findings

Problem / data: How leakage undermines scientific performance claims. Review of reported leakage plus original civil-war-prediction reanalysis.

Architecture / fusion: Compares complex ML approaches with logistic regression; no multimodal fusion.

Training / evaluation: Reproduction and correction of data/analysis errors; classification evaluation.

Relevant result: After corrections, complex models do not substantively outperform the logistic baseline in the case study.

Limits / thesis relevance: The cross-field tally is secondary synthesis; the civil-war reanalysis is original evidence. It illustrates a failure mode, not proof that this thesis has the same leakage.

SQ1 I: model justification | SQ2 D: baseline claims | SQ3 I: deployment distribution | SQ4 D: leakage audit

12. Paper evidence catalogue (continued)

[R27] On Calibration of Modern Neural Networks

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). ICML, PMLR 70.

DOI: No DOI listed in inspected PMLR record.

Source: https://proceedings.mlr.press/v70/guo17a/guo17a.pdf

Access: FT: definitions, protocols and Table 1

Problem / data: Whether predicted confidence matches observed correctness. ImageNet, CIFAR, SVHN and text-classification datasets.

Architecture / fusion: CNNs, deep averaging networks and TreeLSTMs; post-hoc temperature scaling and other calibrators.

Training / evaluation: Fit calibration on validation data; evaluate held-out ECE and other metrics.

Relevant result: Temperature scaling is effective across the studied vision tasks and competitive on text.

Limits / thesis relevance: Mostly in-distribution evidence; ECE depends on binning. A calibrated branch need not identify its reliability after a sensor or population shift.

SQ1 I: encoder confidence | SQ2 I: score combination | SQ3 I: no shift guarantee | SQ4 D: calibration

[R28] Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift

Ovadia, Y., et al. (2019). NeurIPS.

DOI: 10.48550/arXiv.1906.02530 (preprint)

Source: https://arxiv.org/html/1906.02530v2

Access: FT: §4 experiments and conclusions

Problem / data: How predictive uncertainty changes under shift. MNIST, CIFAR-10, ImageNet, text and categorical ad-click tasks.

Architecture / fusion: ResNets, LeNet, LSTM and MLP with ensembles, dropout, variational and calibration alternatives.

Training / evaluation: Validation-tuned models evaluated on shifted/corrupted and out-of-distribution data; multiple metrics.

Relevant result: Ensembles are generally strong; in-distribution temperature calibration does not ensure shifted calibration.

Limits / thesis relevance: Method rankings vary across datasets, and all degrade. This supports uncertainty scrutiny, not the correctness of a particular industrial quality score.

SQ1 I: uncertainty of branches | SQ2 I: ensemble alternative | SQ3 D: shift/noise | SQ4 D: calibration limits

12. Paper evidence catalogue (continued)

[R29] The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets

Saito, T., & Rehmsmeier, M. (2015). PLOS ONE, 10(3), e0118432.

DOI: 10.1371/journal.pone.0118432

Source: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0118432

Access: FT-selected: metric argument and examples

Problem / data: Interpret classifier performance under class imbalance. Synthetic examples and biomedical binary-classification applications.

Architecture / fusion: Metric analysis rather than a new architecture or fusion method.

Training / evaluation: Compare ROC and precision-recall behaviour as class balance and predictions vary.

Relevant result: PR makes poor positive-class precision more visible when ROC can appear reassuring.

Limits / thesis relevance: PR summaries are prevalence-sensitive; the argument does not make ROC useless or establish a deployment threshold. AP is not necessarily identical to trapezoidal PR-AUC.

SQ1 -: no encoder choice | SQ2 I: comparison metric | SQ3 I: condition-specific prevalence | SQ4 D: imbalance/metrics

[R30] Attention is not Explanation

Jain, S., & Wallace, B. C. (2019). NAACL-HLT, 3543-3556.

DOI: 10.18653/v1/N19-1357

Source: https://aclanthology.org/N19-1357/

Access: A+: primary proceedings abstract and paper description

Problem / data: Whether attention weights faithfully explain model predictions. NLP tasks including text classification; full per-task protocol was not extracted here.

Architecture / fusion: Attention-based predictive models; compare attention with feature-importance measures and alternative attention distributions.

Training / evaluation: Correlation, permutation and adversarial-attention analyses.

Relevant result: Different attention patterns can yield similar predictions; attention can weakly agree with other importance measures.

Limits / thesis relevance: Not a theorem about every architecture or every meaning of explanation. Applies indirectly to industrial gates.

SQ1 -: no suitability result | SQ2 I: gate interpretation | SQ3 -: not sensor robustness | SQ4 D: explanation fidelity

12. Paper evidence catalogue (continued)

[R31] Attention is not not Explanation

Wiegreffe, S., & Pinter, Y. (2019). EMNLP-IJCNLP, 11-20.

DOI: 10.18653/v1/D19-1002

Source: https://aclanthology.org/D19-1002/

Access: A+: primary proceedings abstract and methodological description

Problem / data: What conclusions attention interventions justify. Attention-based NLP settings; exact dataset-specific splits not extracted here.

Architecture / fusion: Uniform, multi-seed, frozen-model and adversarial diagnostic comparisons.

Training / evaluation: Re-examines how explanation tests are defined and which model components vary.

Relevant result: Challenges blanket dismissal of attention; explanatory usefulness depends on a specific claim and test.

Limits / thesis relevance: Does not certify industrial gate weights as causal or calibrated. Read with [R30], not as permission to equate attention with trust.

SQ1 -: no encoder ranking | SQ2 I: interpretation of gating | SQ3 -: not corruption proof | SQ4 D: balanced explanation argument

[R32] Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program)

Pineau, J., et al. (2021). JMLR, 22(164), 1-20.

DOI: No DOI listed in inspected JMLR record.

Source: https://jmlr.org/papers/v22/20-303.html

Access: A: primary journal record

Problem / data: Improve transparency and reproducibility of ML findings. NeurIPS 2019 submissions/reproducibility programme, not a prediction dataset.

Architecture / fusion: No model/fusion architecture; code policy, community reproduction challenge and checklist.

Training / evaluation: Programme deployment and analysis; no classifier-training comparison applies.

Relevant result: Documents practical mechanisms for making research inspectable and reproducible.

Limits / thesis relevance: Policy and reporting evidence do not guarantee a specific codebase or metric is correct; no unverified programme effect size is asserted.

SQ1 I: checkpoint provenance | SQ2 I: comparison traceability | SQ3 I: corruption reproducibility | SQ4 D: reproducibility

12. Paper evidence catalogue (continued)

[R33] No Unbiased Estimator of the Variance of K-Fold Cross-Validation

Bengio, Y., & Grandvalet, Y. (2004). JMLR, 5, 1089-1105.

DOI: No DOI listed in inspected JMLR record.

Source: https://jmlr.org/papers/v5/grandvalet04a.html

Access: A+: primary theoretical result and simulation summary

Problem / data: Uncertainty of cross-validation estimates. Mathematical analysis plus numerical learning-problem simulations.

Architecture / fusion: No fusion method; covariance analysis of overlapping CV errors.

Training / evaluation: Eigen-decomposition and simulations varying problem difficulty and fold count.

Relevant result: No universal unbiased estimator of K-fold CV variance exists; naive independence assumptions can underestimate uncertainty.

Limits / thesis relevance: This is a universal impossibility result, not a claim that every interval is invalid. It does not by itself validate or invalidate a particular recording bootstrap.

SQ1 I: representation comparison | SQ2 D: comparison uncertainty | SQ3 I: repeated evaluation | SQ4 D: dependence and intervals

[R34] Strictly Proper Scoring Rules, Prediction, and Estimation

Gneiting, T., & Raftery, A. E. (2007). JASA, 102(477), 359-378.

DOI: 10.1198/016214506000001437

Source: https://sites.stat.washington.edu/raftery/Research/PDF/Gneiting2007jasa.pdf

Access: FT-limited: primary indexed paper; direct fetch timed out

Problem / data: Principled scoring of probabilistic predictions. Mathematical forecast distributions and illustrative applications; no common classification benchmark.

Architecture / fusion: No neural architecture or fusion rule; theory of proper scores.

Training / evaluation: Analytical characterization and examples, not model-training evidence.

Relevant result: Strict propriety makes truthful predictive distributions optimal in expected score.

Limits / thesis relevance: Does not guarantee that a fitted network is calibrated or robust. A proper score measures overall probability quality, not calibration alone.

SQ1 -: no encoder ranking | SQ2 I: probabilistic fusion | SQ3 I: no shift guarantee | SQ4 D: uncertainty evaluation

12. Paper evidence catalogue (continued)

[R35] Multi-sensor and MTConnect Dataset of Metal Cutting Anomaly in Milling from Laboratory and Industry Settings

Kim, E., et al. (2026). Scientific Data. Published 24 April 2026.

DOI: 10.1038/s41597-026-07255-7

Source: https://www.nature.com/articles/s41597-026-07255-7

Access: Limited: publisher-indexed record; full text blocked

Problem / data: Provide milling anomaly data across laboratory and industrial settings. MSM; indexed labels 0 normal, 1 abnormal, 2 tool defect.

Architecture / fusion: Dataset contribution; predictive architecture and fusion protocol not independently verified.

Training / evaluation: Acquisition, annotation and validation details remain unextracted in this review.

Relevant result: Existence and publication metadata verified; no performance result asserted.

Limits / thesis relevance: Cannot yet justify a binary-label mapping, group split, modality availability or transfer claim. Charter descriptions remain project assertions until reconciled with the full source.

SQ1 I: potential sensor source | SQ2 I: possible external task | SQ3 I: external validity only | SQ4 D: unresolved data semantics

[R36] Robust Fault Detection in Mechanical Multimodal Time Series via Self-Supervised Cross-Modal Reconstruction

Jensen, M. M., Hammershøi, D., Wiśniewski, R., & Fink, O. (2026). arXiv preprint, 14 September.

DOI: DOI not independently verified; arXiv identifier 2609.16314.

Source: https://arxiv.org/abs/2609.16314

Access: Screened: primary indexed abstract; full source unavailable

Problem / data: Fault detection with multimodal relationships and operating-condition shift. Three industrial case studies; identities/splits not verified.

Architecture / fusion: Reconstruct each modality from other streams; adaptive test-time anomaly threshold.

Training / evaluation: Self-supervised learning and shifted-condition evaluation described in the abstract.

Relevant result: Authors claim improved robustness; no numerical result is reproduced.

Limits / thesis relevance: Preprint status and unavailable protocol prevent strong weight. Threshold adaptation is a distinct mechanism from quality-conditioned classification.

SQ1 I: learned sensor representations | SQ2 D: reconstruction alternative | SQ3 D: claimed shift/missingness | SQ4 D: test-time information limits